

CLOUDSHIELD AI

03 — Functional Specification

Product-Level Functional Behavior Specification

Field	Value
Document ID	CloudShield AI — 03
Version	1.0
Status	Engineering Baseline
Scope	CloudShield AI product; DAIL treated as an integrated subsystem
Primary Cloud	AWS for MVP
Model Strategy	LLM provider abstraction; hosted model initially, local/open-weight adapter later

AI proposes. Deterministic systems validate. Governance decides. Evidence records. Trusted state is earned.

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1. Purpose and Scope

This Functional Specification defines how CloudShield AI behaves as a product. It converts the product requirements and system architecture into observable workflows, state transitions, decisions, user interactions, service responsibilities, and acceptance behavior.

CloudShield AI is a cloud security and infrastructure intelligence platform. It discovers cloud resources, builds normalized context, evaluates security and configuration posture, identifies and prioritizes findings, uses AI to explain and recommend actions, creates explicit remediation candidates, uses DAIL (Deterministic AI Infrastructure Lifecycle) as a safety and lifecycle subsystem, verifies changes, records evidence, and promotes only validated state to trusted state.

1.1 Functional Boundary

- CloudShield AI is the product boundary.
- DAIL is a bounded subsystem responsible for deterministic lifecycle and safety controls around infrastructure changes.
- AI reasoning is advisory/non-authoritative; it cannot directly become production authority.
- Infrastructure-affecting changes must exist as explicit candidates before verification or promotion.
- AWS is the primary cloud provider for the MVP; provider-specific integrations are isolated behind adapters.
- The initial LLM is accessed through an AI abstraction layer so the model/provider can be changed later.

1.2 Functional Design Rule

Every material product action should be explainable through an explicit chain: observed context →analysis → finding/risk →recommendation →candidate →impact →verification →governance decision →promotion →evidence.

2. Functional Model

The product is organized into the following functional domains:

Domain	Primary Functional Responsibility
Cloud Connectivity	Authenticate, validate access, enumerate accounts/regions and maintain cloud connection state.
Discovery & Inventory	Discover resources, normalize them into product assets, reconcile repeated observations.
Configuration Intelligence	Capture configuration snapshots, compare versions and identify drift.
Security Analysis	Evaluate rules/policies and create actionable findings.
Risk Intelligence	Calculate and prioritize risk using deterministic/contextual inputs.
Dependency & Impact	Build relationships and determine affected resources and invalidation scope.
AI Intelligence	Assemble context, explain findings, recommend remediation and generate structured proposals.
Remediation	Represent proposed infrastructure changes as explicit candidates.
DAIL	Identity, lifecycle, dependency/impact gates, verification and promotion controls.
Governance	Apply policies, approvals, authorization and human decisions.
Evidence & Audit	Record decisions, artifacts, verification results and provenance.
Monitoring & Reporting	Surface current posture, changes, failures, trends and reports.

2.1 Canonical Functional Loop

DISCOVER →INVENTORY →CONFIGURE →ANALYZE →FINDING →RISK →AI EXPLANATION/RECOMMENDATION →
REMEDATION CANDIDATE →DAIL IMPACT →VERIFY →DECIDE →PROMOTE →TRUSTED STATE →MONITOR

3. Actors, Roles and Permissions

Actor	Core Capabilities	Restrictions
Platform Administrator	Manage platform configuration, integrations, policies and operational settings.	Must not bypass audit/evidence requirements.
Security Analyst	Review findings, risk, dependencies, AI explanations and remediation candidates.	Cannot promote without required authority/policy.
Infrastructure Engineer	Inspect assets, review candidates, execute or approve infrastructure changes.	Production changes remain subject to verification/governance.
Approver	Approve/reject eligible candidates.	Approval is limited to authorized scope.
Auditor	Inspect evidence, history, decisions and verification records.	Read-only.
Automation/Worker	Run discovery, analysis, AI, verification and evidence jobs.	No unrestricted human-equivalent authority.

3.1 Authorization Behavior

- Every protected operation is evaluated against authenticated identity, tenant/account scope, role and resource scope.
- Read operations and mutation operations are separately authorized.
- Promotion authority is distinct from candidate creation authority.
- System workers use service identities with least privilege.
- A failed authorization must not partially execute the requested mutation.
- All material authorization-sensitive actions are auditable.

4. Product Functional Areas

4.1 Cloud Connectivity

CF-CONN-001: The system shall allow an authorized user to register an AWS account/cloud connection.

CF-CONN-002: The system shall validate credentials/assumed-role access before marking a connection usable.

CF-CONN-003: The system shall expose connection status, last validation time and validation errors.

CF-CONN-004: The system shall not store or expose unnecessary long-lived cloud secrets.

CF-CONN-005: A disabled or invalid connection shall prevent new discovery jobs from starting against that connection.

4.2 Discovery and Inventory

CF-DISC-001: An authorized user or scheduled job shall initiate discovery for a configured AWS scope.

CF-DISC-002: Discovery shall enumerate supported resources and collect required configuration attributes.

CF-DISC-003: The system shall normalize provider-specific objects into canonical CloudShield AI asset representations.

CF-DISC-004: Repeated observations of the same logical asset shall reconcile to a stable asset identity where deterministically possible.

CF-DISC-005: Discovery shall record provenance, timestamp and source scope for collected observations.

CF-DISC-006: Partial discovery failures shall be represented explicitly and shall not silently appear as complete inventory.

4.3 Configuration Snapshots

CF-CONF-001: The system shall persist point-in-time configuration snapshots for supported assets.

CF-CONF-002: The system shall calculate configuration differences between comparable snapshots.

CF-CONF-003: The system shall distinguish observed configuration from proposed configuration.

CF-CONF-004: A proposed configuration shall not be represented as trusted until successful verification and promotion.

5. Cloud Onboarding Workflow

Functional sequence:

29. User creates a cloud connection and selects account/tenant scope.
30. System validates credentials and required permissions.
31. System records connection metadata and validation evidence.
32. System initializes discovery scope and supported service configuration.
33. User starts discovery or an authorized scheduler starts it.
34. Discovery workers collect cloud observations asynchronously.
35. Collected resources are normalized and reconciled.
36. Configuration snapshots and discovery evidence are persisted.
37. Security analysis is queued for eligible assets.
38. Dashboard updates the connection, inventory and analysis status.

5.1 Onboarding States

State	Meaning
CONFIGURED	Connection configuration exists but validation is not complete.
VALIDATING	Credentials/permissions are being checked.
READY	Connection is usable for configured operations.
DEGRADED	Connection works partially or required scopes are unavailable.
DISABLED	Connection is intentionally unavailable.
ERROR	Validation or operational failure prevents intended use.

6. Discovery, Inventory and Normalization

6.1 Discovery Functional Behavior

- Discovery is job-oriented and asynchronous.
- Each job has a scope, start time, status, progress metadata, source account and execution identity.
- Workers retrieve provider data and convert it into canonical observations.
- Unsupported resources are retained as discoverable metadata when useful, but are not evaluated as supported security objects.
- Transient provider errors are retried according to bounded retry policy.
- Permanent/permission failures are recorded with enough context for remediation.

6.2 Asset Reconciliation

- Provider identifiers are preferred as stable identity anchors.
- Normalization removes provider-specific representation differences where possible.
- An observation may update an existing asset, create a new asset, or mark a previous asset as no longer observed.
- Identity conflicts must be surfaced rather than merged heuristically.
- Asset history remains available for audit and drift analysis.

7. Configuration Snapshot and Drift

CloudShield AI treats configuration as a versioned observation rather than a single mutable value. This allows the system to distinguish what the cloud currently reports from what a candidate proposes.

Comparison	Functional Meaning
Observed →Observed	Detect environmental/configuration drift between cloud observations.
Observed →Proposed	Show what a candidate would change.
Proposed →Verified	Show whether the candidate produced the intended result.
Verified →Trusted	Record the promoted configuration as accepted product state.

CF-DRIFT-001: The system shall detect material differences between relevant observations.

CF-DRIFT-002: Drift events shall be attributable to an observation and timestamp.

CF-DRIFT-003: Drift shall be eligible to trigger re-analysis where configured.

CF-DRIFT-004: A detected drift shall not itself authorize remediation.

8. Security Analysis and Findings

8.1 Rule Evaluation

- Rules evaluate normalized asset/configuration context against deterministic security conditions.
- A rule evaluation produces pass, fail, not-applicable or evaluation-error semantics.
- Evaluation errors are not silently converted to passes.
- Each finding references the rule, affected asset(s), observed evidence and evaluation time.

8.2 Finding Lifecycle

State	Meaning
OPEN	Finding currently exists and requires attention.
ACKNOWLEDGED	A user has reviewed it but it remains unresolved.
IN_REMEDIATION	A remediation candidate is associated with the finding.
VERIFIED	The intended corrective state has been verified.
RESOLVED	The product recognizes the finding as resolved.
REOPENED	Previously resolved condition has returned.
SUPPRESSED	Finding is intentionally hidden by authorized policy with reason/evidence.

CF-FIND-001: Every actionable finding shall include severity/risk context, affected asset(s), evidence and remediation guidance where available.

CF-FIND-002: Finding state transitions shall be auditable.

CF-FIND-003: Suppression shall require an authorized reason and shall not erase historical evidence.

9. Risk Intelligence and Prioritization

Risk is a product-level prioritization signal. It combines deterministic security findings with contextual information such as asset criticality, exposure, dependency relationships and available evidence. The exact scoring implementation may evolve, but inputs and score versions must remain traceable.

Risk Input	Functional Role
Finding severity	Base security significance.
Asset criticality	Business/operational importance.
Exposure	External or attack-surface relevance.
Dependency impact	Potential blast radius.
Evidence confidence	Confidence in the observed condition.
Compensating controls	Context that may reduce effective risk.
Score version	Makes risk calculations reproducible.

CF-RISK-001: Risk results shall identify the score/calculation version.

CF-RISK-002: Users shall be able to inspect the primary factors contributing to prioritization.

CF-RISK-003: Risk prioritization shall not silently replace underlying findings or evidence.

10. Dependency and Impact Analysis

CloudShield AI represents infrastructure relationships as a dependency graph. Impact analysis determines which resources, findings, candidates and trusted assumptions may be affected by a change.

CF-DEP-001: The system shall represent supported resource relationships explicitly.

CF-DEP-002: A remediation candidate shall reference the assets it intends to affect.

CF-IMP-001: The system shall calculate affected resources using deterministic dependency traversal.

CF-IMP-002: Impact results shall distinguish directly affected assets from transitively affected assets.

CF-IMP-003: A candidate shall be blocked or escalated when impact analysis identifies a policy-defined unsafe condition.

10.1 Impact Output

- Direct targets
- Dependent resources
- Potentially invalidated findings
- Potentially invalidated verification results
- Trusted-state assumptions requiring revalidation
- Blast-radius summary
- Evidence references used to derive the result

11. AI Intelligence Workflow

AI provides contextual reasoning and recommendations. It does not directly control infrastructure. The functional interface to AI is structured and provider-independent.

11.1 Context Assembly

39. Identify the task type: explanation, prioritization support, remediation recommendation, or other supported operation.
40. Collect only authorized relevant context: findings, assets, configuration, dependencies, evidence and policy constraints.
41. Construct a bounded structured context package.
42. Invoke the configured model provider through the AI abstraction layer.
43. Validate the response against the expected structured schema.
44. Attach model/provider metadata and context provenance.
45. Return the AI result as advisory product data.

CF-AI-001: AI requests shall use a defined structured contract.

CF-AI-002: AI output shall be schema-validated before downstream use.

CF-AI-003: AI output shall not directly receive unrestricted production credentials.

CF-AI-004: AI-generated infrastructure changes shall become explicit candidates before verification or execution.

CF-AI-005: The system shall retain model/provider metadata sufficient to reproduce or audit the AI decision context where policy permits.

CF-AI-006: The AI layer shall support provider replacement without changing core product workflows.

12. Remediation and Change Candidates

A remediation candidate is the product's explicit representation of an intended infrastructure change. It may originate from AI, a user, a rule/template, or another authorized source.

Candidate State	Functional Meaning
DRAFT	Candidate is being constructed.
READY_FOR_IMPACT	Required fields are complete and candidate can be analyzed.
IMPACTED	Dependency/impact analysis has completed.

AWAITING_VERIFICATION	Candidate is ready for deterministic verification workflow.
VERIFYING	Verification is executing.
VERIFIED	Verification gates passed.
APPROVAL_REQUIRED	Governance requires an authorized decision.
APPROVED	Required approval has been recorded.
PROMOTING	Promotion is executing.
PROMOTED	Trusted state was updated.
REJECTED	Candidate was rejected.
BLOCKED	Safety/policy gate prevents progression.
FAILED	Execution or verification failed.

CF-CAND-001: Every candidate shall have a unique immutable identity.

CF-CAND-002: Candidate content shall be versioned or content-addressable so the verified object is unambiguous.

CF-CAND-003: A candidate shall declare intended targets and expected post-change conditions.

CF-CAND-004: Rejected, blocked and failed candidates shall remain auditable.

CF-CAND-005: A candidate shall not be promoted solely because AI recommended it.

13. DAIL Integration Workflow

DAIL is the deterministic lifecycle subsystem inside CloudShield AI. The product integrates with DAIL through explicit contracts rather than embedding lifecycle logic throughout unrelated product modules.

13.1 Integration Sequence

46. CloudShield AI creates or receives a remediation candidate.
47. Candidate is assigned DAIL identity/lifecycle metadata.
48. DAIL resolves relevant dependencies and expected state relationships.
49. DAIL performs impact/invalidation analysis.
50. Candidate enters verification workflow.
51. Verification produces structured pass/fail results and evidence.
52. Governance evaluates approval/policy requirements.
53. Eligible candidate is promoted.
54. DAIL records lifecycle evidence and trusted-state transition.
55. CloudShield AI refreshes findings, risk, inventory and dashboard state.

13.2 DAIL Boundary

- CloudShield AI owns product workflows and user-facing orchestration.
- DAIL owns deterministic lifecycle controls for infrastructure change candidates.
- The AI layer does not bypass DAIL.
- DAIL does not become the product's entire security/risk/intelligence layer.
- Evidence generated by DAIL is consumed as first-class product evidence.

14. Verification, Approval and Promotion

14.1 Verification

- Verification compares intended post-change conditions with observed results.
- Verification can include configuration assertions, security invariants, dependency expectations and provider observations.
- Verification must be tied to the exact candidate identity/version.
- A verification failure blocks promotion unless a separately defined emergency/governance path exists.
- Verification output includes pass/fail status, checks executed, observations, timestamps and evidence.

14.2 Governance Decision

Decision	Behavior
APPROVE	Candidate may proceed if all technical gates pass.
REJECT	Candidate terminates and remains auditable.
REQUEST_CHANGES	Candidate returns to an editable state with rationale.
ESCALATE	Candidate requires a different authority or policy decision.
AUTO_APPROVE	Only allowed when explicit policy permits and technical gates pass.

14.3 Promotion

CF-PROM-001: Promotion shall require successful verification and all required governance gates.

CF-PROM-002: Promotion shall update trusted state only after successful completion.

CF-PROM-003: Promotion failures shall leave trusted state unchanged.

CF-PROM-004: Promotion shall produce an immutable audit/evidence record.

15. Trusted State and Continuous Monitoring

Trusted state represents product state that has earned acceptance through successful verification and promotion. It is not synonymous with the latest observed cloud state.

State Layer	Meaning
OBSERVED	What the cloud/provider currently reports.
PROPOSED	What a candidate intends to produce.
VERIFIED	What the system has confirmed after candidate evaluation/execution.
TRUSTED	What governance has accepted as the current trusted product state.

CF-TRUST-001: The product shall maintain explicit distinction between observed, proposed, verified and trusted state.

CF-TRUST-002: A new observation may invalidate trusted assumptions when configured conditions are met.

CF-MON-001: Scheduled or event-triggered monitoring shall be able to initiate discovery/re-analysis.

CF-MON-002: Material drift shall update relevant findings and risk context.

CF-MON-003: Monitoring shall not silently mutate infrastructure.

16. Evidence, Audit and Reporting

Evidence is a first-class functional output. The product must be able to explain what was observed, what was recommended, what changed, what was verified, who/what decided, and why trusted state changed.

Evidence Category	Examples
Observation	Cloud API response metadata, configuration snapshot, discovery timestamp.
Analysis	Rule evaluation, risk calculation, dependency traversal.
AI	Structured request/response metadata, model/provider and context references.
Candidate	Candidate content/version, target resources, expected state.
Verification	Checks, results, observed post-change values.
Governance	Approval/rejection/escalation and policy basis.
Promotion	Trusted-state transition and resulting observation.
Operational	Job execution, retries, failures and worker identity.

CF-EVID-001: Material state transitions shall create traceable evidence.

CF-EVID-002: Evidence shall be linked to the relevant asset, finding, candidate or job.

CF-EVID-003: Audit records shall identify actor/service identity and timestamp.

CF-REP-001: The product shall provide current posture and historical reporting appropriate to authorized users.

17. Dashboard and User Experience Behavior

17.1 Core Dashboard

- Cloud connection health and discovery freshness.
- Security posture summary and finding counts.
- Risk-prioritized assets/findings.
- Open remediation candidates and their lifecycle states.
- Verification failures and blocked candidates.
- Recent drift and trusted-state changes.
- Evidence/audit navigation.

17.2 Finding Detail

- Finding description and severity.
- Affected resource(s).
- Evidence and observed configuration.
- Risk factors.
- Dependency/impact context.
- AI explanation/recommendation when available.
- Related remediation candidates.
- Verification and resolution history.

17.3 Candidate Detail

- Candidate identity/version.
- Origin: human, AI, rule/template or automation.

- Target resources.
- Proposed changes.
- Expected post-change state.
- Impact analysis.
- Verification status and evidence.
- Governance decisions.
- Promotion/trusted-state result.

18. Experimentation and Evaluation

CloudShield AI may evaluate alternative AI prompts, models, policies, risk heuristics or remediation strategies. Experiments must remain isolated from trusted production state.

CF-EXP-001: Experiments shall have explicit identity, configuration and evaluation criteria.

CF-EXP-002: Experiment outputs shall be distinguishable from production decisions.

CF-EXP-003: Experimental infrastructure changes shall not become trusted production state without the normal candidate, verification and governance path.

CF-EXP-004: AI evaluation shall measure structured quality and safety characteristics, not only text quality.

19. Asynchronous Jobs and Event Processing

Long-running discovery, analysis, AI, verification, reporting and monitoring operations shall execute asynchronously where appropriate. The user-facing product should expose job status rather than blocking on long-running operations.

Job State	Meaning
QUEUED	Accepted but not started.
RUNNING	Worker is processing.
SUCCEEDED	Completed successfully.
PARTIAL	Completed with explicitly recorded partial failures.
FAILED	Terminal failure.
CANCELLED	Stopped by authorized action/system policy.

CF-JOB-001: Jobs shall have unique identity and correlation metadata.

CF-JOB-002: Retries shall be bounded and idempotency-aware.

CF-JOB-003: Duplicate delivery shall not create duplicate irreversible state transitions.

CF-JOB-004: Job failures shall be visible to authorized users and operational tooling.

20. Error and Exception Handling

Failure Class	Expected Functional Response
Authentication failure	Reject operation; record security-relevant event.
Authorization failure	Reject operation without partial mutation.
Cloud API transient failure	Retry within bounded policy; expose status if unresolved.
Cloud API permission failure	Mark affected operation degraded/failed with actionable

	evidence.
Invalid AI output	Reject/repair through bounded validation path; never treat malformed output as authority.
Dependency analysis failure	Block candidate progression where impact is required.
Verification failure	Block promotion; retain failed evidence.
Approval rejection	Terminate candidate progression and retain decision.
Promotion failure	Do not advance trusted state; record failure.
Storage failure	Fail closed for operations requiring durable evidence.

Safety-critical ambiguity should default toward non-promotion rather than optimistic continuation.

21. API-Facing Functional Behavior

The detailed API contract is specified separately, but product behavior requires the following API semantics:

- Authenticated requests must be scoped to an authorized tenant/account/resource boundary.
- Mutation endpoints must validate input before starting side effects.
- Long-running operations return job/correlation identifiers rather than requiring synchronous completion.
- Candidate APIs expose immutable identity and lifecycle state.
- Verification and promotion operations must enforce state preconditions.
- Responses should distinguish current state from historical evidence.
- Idempotency is required for operations that can trigger irreversible side effects.
- API errors should expose actionable categories without leaking secrets or sensitive provider credentials.

22. Functional State Models

22.1 Candidate State Transition Rules

From	Trigger	To	Gate
DRAFT	Complete required fields	READY_FOR_IMPACT	Schema/authorization validation
READY_FOR_IMPACT	Impact analysis	IMPACTED	Dependency graph available
IMPACTED	Proceed	AWAITING_VERIFICATION	Impact policy passes
AWAITING_VERIFICATION	Start verification	VERIFYING	Candidate immutable
VERIFYING	All checks pass	VERIFIED	Verification success
VERIFYING	Any blocking check fails	FAILED	Fail closed
VERIFIED	Governance required	APPROVAL_REQUIRED	Policy
VERIFIED	Auto approval permitted	APPROVED	Policy + technical gates
APPROVAL_REQUIRED	Approve	APPROVED	Authorized decision
APPROVAL_REQUIRED	Reject	REJECTED	Authorized decision
APPROVED	Promote	PROMOTING	Preconditions rechecked
PROMOTING	Success	PROMOTED	Post-change confirmation
PROMOTING	Failure	FAILED	Trusted state unchanged

22.2 Finding State Rules

From	Trigger	To
OPEN	User review	ACKNOWLEDGED
OPEN/ACKNOWLEDGED	Candidate created	IN_REMEDIATION
IN_REMEDIATION	Successful verification	VERIFIED
VERIFIED	Resolution recognized	RESOLVED
RESOLVED	Condition returns	REOPENED
OPEN/ACKNOWLEDGED	Authorized suppression	SUPPRESSED

SUPPRESSED	Suppression expires/removal	OPEN
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23. End-to-End Use Cases

UC-01 — Discover and Analyze an AWS Environment

56. Administrator configures and validates an AWS connection.
 57. User starts discovery for selected account/regions.
 58. Discovery job collects resources and configuration.
 59. Assets are normalized/reconciled.
 60. Security analysis evaluates supported rules.
 61. Findings and risk are calculated.
 62. Dashboard presents posture and evidence.
- Success: inventory and findings are available with provenance and job history.

Failure: incomplete scopes/errors are explicitly shown; the system does not claim complete posture.

UC-02 — AI-Assisted Remediation

63. User opens a finding.
 64. System assembles authorized context.
 65. AI produces structured explanation and remediation recommendation.
 66. System validates the AI output.
 67. User/system creates an explicit candidate from the recommendation.
 68. Candidate undergoes DAIL identity, dependency and impact processing.
 69. Verification executes.
 70. Governance decision is recorded.
 71. If approved and all gates pass, promotion occurs.
 72. System refreshes observation and finding state and records evidence.
- Success: intended change is verified and trusted state is updated.

Failure: candidate remains blocked/failed/rejected and trusted state is not advanced.

UC-03 — Detect Drift After Promotion

73. Monitoring triggers discovery.
74. Observed configuration differs from trusted state.
75. System records drift.
76. Relevant findings/risk are recalculated.
77. Existing assumptions or verification evidence may be invalidated according to policy.
78. Dashboard surfaces the new condition.

UC-04 — Candidate Rejected by Governance

79. Candidate passes technical analysis and reaches approval.
80. Authorized approver reviews impact, evidence and expected state.
81. Approver rejects candidate with reason.
82. Candidate becomes REJECTED.
83. No promotion occurs.
84. Evidence records the decision and candidate remains available for audit.

24. MVP Functional Slice

The first implementation should prove the complete control loop with limited scope rather than implement every cloud service.

- AWS connection onboarding and validation.
- Discovery for a focused set of AWS resource types.
- Canonical asset inventory.
- Configuration snapshots and basic drift detection.
- Deterministic security rules and findings.
- Risk prioritization.
- Basic dependency graph.
- AI explanation and remediation recommendation through provider abstraction.
- Explicit remediation candidate creation.
- DAIL identity/dependency/impact integration.
- Deterministic verification for a constrained set of change types.
- Human approval.
- Promotion to trusted state.
- Evidence/audit trail.
- Basic dashboard and job status.

The MVP should demonstrate that the full product loop works safely end-to-end. Breadth of cloud services is secondary to correctness of the control loop.

25. Functional Acceptance Criteria

ACC-01: A valid AWS connection can be configured, validated and observed through its lifecycle.

ACC-02: Discovery creates normalized assets with provenance and explicit partial-failure reporting.

ACC-03: Configuration snapshots can be compared and material drift is detectable.

ACC-04: Security rules create traceable findings tied to evidence.

ACC-05: Risk prioritization is reproducible for a given score version and inputs.

ACC-06: Dependency/impact analysis identifies affected resources for supported candidates.

ACC-07: AI responses are schema-validated and remain advisory.

ACC-08: AI cannot directly mutate production infrastructure.

ACC-09: Every infrastructure-affecting proposal becomes an explicit candidate.

ACC-10: Candidate lifecycle states and preconditions are enforced.

ACC-11: DAIL participates in candidate identity, impact, verification and promotion controls.

ACC-12: Verification failures block promotion.

ACC-13: Governance decisions are explicit and auditable.

ACC-14: Successful promotion updates trusted state only after required gates pass.

ACC-15: Promotion failure leaves trusted state unchanged.

ACC-16: Drift can invalidate relevant assumptions and trigger re-analysis.

ACC-17: Evidence connects observations, analysis, AI output, candidate, verification, decision and promotion.

ACC-18: Long-running operations expose job state and correlation identifiers.

ACC-19: The system fails closed when safety-critical validation/evidence is unavailable.

ACC-20: The product can replace the initial LLM provider through the AI abstraction boundary.

26. Traceability to SRS and Architecture

Functional Area	SRS Alignment	Architecture Alignment
Cloud onboarding	Cloud integration, security, reliability	Cloud adapter + application layer
Discovery/inventory	Asset discovery and data requirements	Discovery workers + relational system of record
Security/risk	Findings, risk, analysis	Security/risk modules + async workers
AI intelligence	AI abstraction and structured outputs	AI service/worker + provider adapter
Remediation	Candidate-based change model	Application + DAIL boundary
DAIL	Lifecycle/verification/promotion requirements	Dedicated bounded subsystem
Evidence	Audit/evidence requirements	Evidence service/storage + immutable records
Monitoring	Observability/reliability	Jobs/events/workers + monitoring
Dashboard	UX/product behavior	Web UI + API/application layer
Experiments	Evaluation requirements	Isolated experiment/evaluation workflow

This specification intentionally operationalizes the product-level SRS and architecture without redefining them. Lower-level specifications should elaborate implementation contracts while preserving these functional boundaries.

27. Functional Constraints and Non-Goals

- CloudShield AI is not an autonomous unrestricted cloud operator.
- The AI model is not the system of record.
- AI output is not trusted merely because it is syntactically valid.
- DAIL is not the entire CloudShield AI product.
- MVP does not require training a foundation model from scratch.
- MVP does not require a proprietary local foundation model.
- The system does not treat observed cloud state as automatically trusted product state.
- Security findings do not automatically authorize remediation.
- The platform should avoid premature microservice decomposition where a modular monolith/workers are sufficient.
- Exact provider-specific API schemas, Terraform implementation details and deployment manifests belong in lower-level technical specifications.

28. Glossary

Term	Definition
CloudShield AI	The complete cloud security and infrastructure intelligence product.
DAIL	Deterministic AI Infrastructure Lifecycle; a subsystem for deterministic infrastructure-change lifecycle and safety controls.
Asset	Normalized representation of a cloud resource or relevant infrastructure object.
Observation	Provider-reported state captured by CloudShield AI.
Finding	A security/configuration condition identified by analysis.
Risk	Contextual prioritization signal derived from findings and environment context.
Candidate	Explicit representation of an intended infrastructure change.
Dependency Graph	Graph representing supported relationships among infrastructure assets.
Impact Analysis	Deterministic evaluation of resources and assumptions affected by a candidate.
Verification	Deterministic evaluation that intended post-change conditions hold.
Promotion	Governed transition of verified state into trusted state.
Trusted State	Product state accepted after required verification and governance.
Evidence	Recorded artifacts supporting observations, analysis, decisions and state transitions.
AI Abstraction Layer	Provider-independent interface for model invocation and structured AI results.

Document Completion Statement

CloudShield AI Functional Specification v1.0 establishes the product-level functional contract required to move from architectural design toward implementation. It preserves the core separation between product capabilities and the DAIL subsystem while defining the complete discover-to-trusted-state control loop.

Next planned document: 04 — CloudShield AI Product & UX Specification.